



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
19-07-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

PAPER-1

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

-
1. Two coaxial coils A (12500 turns) and B (16000 turns) are placed such that 60% of flux produced in A passes through coil B. It is found that a current of 5A in A produces a flux of 0.6 mWb through its cross section and same current in B produces a flux of 0.8 mWb through its cross section. The coefficient of mutual induction of the pair of coils :
- (A) 1.152 H (B) 2.1 H (C) 0.5 H (D) 3 H
-

Space for Rough Work

2. A long non conducting cylinder of radius a carries a uniform surface charge density σ coul/m² rotates about its axis with an angular speed ω . The magnetic energy stored in cylinder per unit of its length (U) is :-

(A) $U = \frac{1}{3} \pi \mu_0 \sigma^2 a^4 \omega^2$

(B) $U = \frac{1}{2} \pi \mu_0 \sigma^2 a^4 \omega^2$

(C) $U = \frac{1}{4} \pi \mu_0 \sigma^2 a^4 \omega^2$

(D) $U = \pi \mu_0 \sigma^2 a^4 \omega^2$

3. A single loop circuit consists of a battery of emf E , a resistance R and a coil of inductance L connected in series. The circuit is in steady state and at $t = 0$ suddenly the coil inductance is increased by n times by insertion of an iron rod in it. If the current in circuit as a function of

time is $i = \frac{E}{R} \left[1 - \left(1 - \frac{1}{n} \right) e^{-\frac{\alpha R t}{nL}} \right]$, then the value of α is

(A) 2

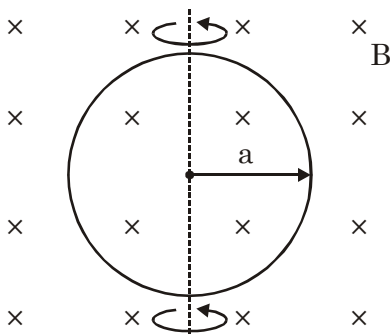
(B) 3

(C) 1

(D) 4

Space for Rough Work

4. Figure shows a circular coil of radius a and resistance R placed in a uniform magnetic induction B . If the coil is rotated about its diametrical axis by 180° , the amount of charge flown through the coil is :



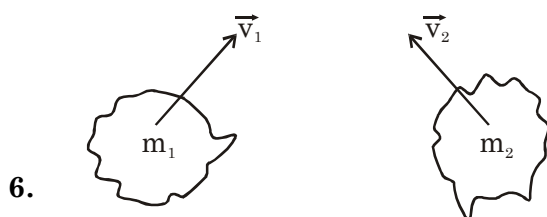
- (A) $\Delta q = \frac{2B\pi a^2}{R}$ (B) $\Delta q = \frac{3B\pi a^2}{R}$ (C) $\Delta q = \frac{4B\pi a^2}{R}$ (D) $\Delta q = \frac{B\pi a^2}{R}$
5. Find magnetic field due to current I flowing in an elliptical loop at its focus. Equation of ellipse (in polar co-ordinates as shown) is $\frac{\ell}{r} = (1 + e \cos \theta)$. Here e is eccentricity which is a constant. ℓ is also a constant :

- (A) $\frac{\mu_0 I}{3\ell}$ (B) $\frac{\mu_0 I}{\ell}$ (C) $\frac{3\mu_0 I}{2\ell}$ (D) $\frac{\mu_0 I}{2\ell}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.



Non-uniform bodies

Consider the diagram above. The two non-uniform bodies shown above undergo a perfectly inelastic collision. After the collision, the bodies stick to each other. Find:

1. The final velocity of the system $\vec{V}_f$.
2. The unit vector along the line of impact $\hat{n}$.

(A) $\vec{V}_f = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2}$

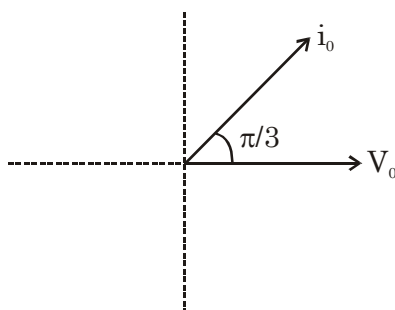
(B) $\vec{V}_f = \vec{v}_1 + \vec{v}_2$

(C) $\hat{n} = \pm \frac{m_1 m_2}{m_1 + m_2} (\vec{v}_2 - \vec{v}_1)$

(D) $\hat{n} = \pm \frac{(\vec{v}_2 - \vec{v}_1)}{|\vec{v}_2 - \vec{v}_1|}$

Space for Rough Work

7. Suppose we have a stationary and fixed "Superconducting" ring of radius "R" and Self Inductance "L" placed on floor in the X-Y plane. There is a uniform external magnetic field in the Negative Z direction, varying with time as $\mathbf{B} = K\mathbf{t}$, where "K" is a positive constant and "t" is time. Tick the correct options below.
- (A) The Net Induced EMF in the ring will be zero, but Net Current in the ring will be Non-Zero.
- (B) The Net Induced EMF in the ring will be Non-Zero and the Net Current in it will also be Non-Zero.
- (C) The induced current in the ring will be equal to $= \frac{Kt\pi R^2}{L}$
- (D) The induced current in the ring will be equal to $= \frac{K\pi R^2}{L}$
8. For an LCR series circuit, phasors of current i and applied voltage $V = V_0 \sin \omega t$ are shown in diagram at $t = 0$. Which of the following is/are **CORRECT** ?



- (A) At $t = \frac{\pi}{2\omega}$, instantaneous power supplied by source is negative.
- (B) From $0 < t < \frac{2\pi}{3\omega}$, average power supplied by source is positive.
- (C) At $t = \frac{5\pi}{6\omega}$, instantaneous power supplied by source is negative.
- (D) If ω is increased slightly, angle between the two phasor decreases.

Space for Rough Work

9. Following are equations of four waves :

$$(i) \ y_1 = a \sin \omega \left(t - \frac{x}{v} \right)$$

$$(ii) \ y_2 = a \cos \omega \left(t + \frac{x}{v} \right)$$

$$(iii) \ z_2 = a \sin \omega \left(t - \frac{x}{v} \right)$$

$$(iv) \ z_2 = a \cos \omega \left(t + \frac{x}{v} \right)$$

Which of the following statements is/are **CORRECT** ?

- (A) On superposition of waves (i) and (iii), a travelling wave having amplitude $a\sqrt{2}$ will be formed.
- (B) Superposition of waves (ii) and (iii) is not possible.
- (C) On superposition of wave (i) and (ii), a transverse stationary wave having maximum amplitude $a\sqrt{2}$ will be formed.
- (D) On superposition of waves (iii) and (iv), a transverse stationary wave will be formed.
10. A progressive wave having amplitude 5 m and wavelength 3m. If the maximum average velocity of particle in half time period is 5 m/s and wave is moving in the positive x-direction then find which may be the **CORRECT** equation(s) of the wave ? [where x in meter]

$$(A) \ 5 \sin \left(\frac{2\pi}{5} t - \frac{2\pi}{3} x \right)$$

$$(B) \ 4 \sin \left(\frac{\pi t}{2} - \frac{2\pi}{3} x \right) + 3 \cos \left(\frac{\pi t}{2} - \frac{2\pi}{3} x \right)$$

$$(C) \ 5 \sin \left(\frac{\pi t}{2} - \frac{2\pi}{3} x \right)$$

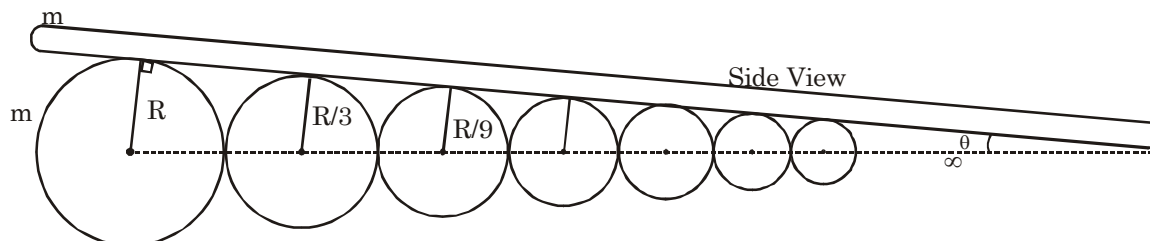
$$(D) \ 3 \cos \left(\frac{2\pi}{5} t - \frac{2\pi}{3} x \right) - 4 \sin \left(\frac{2\pi}{5} t - \frac{2\pi}{3} x \right)$$

Space for Rough Work

11. A standing wave of time period T is set up in a string clamped between two rigid supports. At $t = 0$ antinode is at its maximum displacement $2A$.
- (A) The energy density of a node is equal to energy density of an antinode for the first time at $t = T/4$.
- (B) The energy density of node and antinode becomes equal after $T/2$ second.
- (C) The displacement of the particle from the mean position at antinode at $t = \frac{T}{8}$ is $\sqrt{2}A$.
- (D) The displacement of the particle from the mean position at node is zero
12. A particle of specific charge $\frac{q}{m} = \pi \times 10^{10} \text{ Ckg}^{-1}$ is projected from the origin along the positive x-axis with a velocity of 10^5 ms^{-1} in a uniform magnetic field $\vec{B} = -2 \times 10^{-3} \hat{k}$ tesla. Choose **CORRECT** alternative(s) :
- (A) The centre of the circle lies on the y-axis
- (B) The time period of revolution is 10^{-7} s .
- (C) The radius of the circular path is $\frac{5}{\pi} \text{ mm}$.
- (D) The velocity of the particle at $t = \frac{1}{4} \times 10^{-7} \text{ s}$ is $10^5 \hat{j} \text{ m/s}$

Space for Rough Work

13. In the following arrangement there are infinite discs with decreasing radii as shown. There is infinitesimal gap between consecutive discs. Largest disc has radius R and the radius of each successive disc is $1/3^{\text{rd}}$ of the radius of the disc to its left. The mass of the largest disc is m .



Infinite number of discs of same mass density pivoted at their centers.

A stick of mass m (same as the mass of the largest disc) is placed on top of the discs (touching them all) and is released. Assuming that the stick doesn't slip. Its tangential acceleration (a_t) and the friction experienced by it (f_r) are :

(A) $a_t = \frac{8g}{25}$

(B) $a_t = \frac{3g}{20}$

(C) $f_r = \frac{9mg}{50}$

(D) $f_r = \frac{7g}{25}$

Space for Rough Work

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

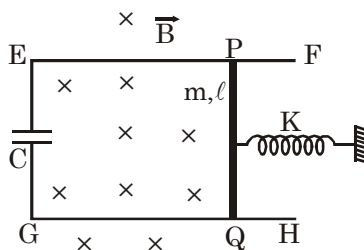
		+		-	
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

		+		-	
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

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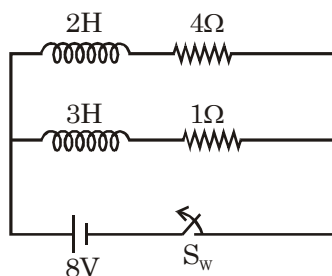
- Figure shows a capacitor C, connected to two straight smooth wire EF and GH. A rod PQ of mass m and length ℓ is placed on wires which connected to one end of a spring of force constant k of which other end is fixed. Whole setup is in a uniform magnetic field B. If the time period

of small oscillation of rod PQ is $2\pi\sqrt{\frac{m + \alpha CB^2 \ell^2}{k}}$, then the value of α is

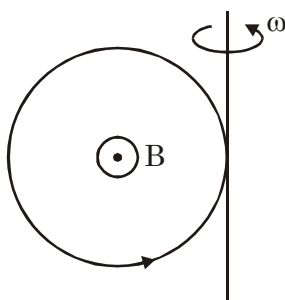


Space for Rough Work

2. Switch was initially closed in the following arrangement till the steady state is reached in the circuit. Now the switch is suddenly opened. Current (in Amp.) in the remaining circuit immediately after opening the switch is :



3. A current carrying ring, carrying a constant current $\frac{2}{\pi}$ Amp., radius 1m, mass $\frac{2}{3}$ kg and having 10 windings is free to rotate about its tangential vertical axis. A uniform magnetic field of 1 tesla is applied perpendicular to its plane. How much minimum angular velocity (in rad/sec) should be given to the ring in the direction shown, so that it can rotate 270° in that direction. Write your answer in nearest single digit in rad/sec.



Space for Rough Work

4. A solenoid with inductance $L = 7 \text{ mH}$ and active resistance $R = 44 \Omega$ is first connected to a source of direct voltage V_0 and then to a source of sinusoidal voltage with effective value $V = V_0$. The frequency (in kilo Hertz) of the oscillator at which the power consumed by the solenoid be $\eta = 5.0$ times less than in the former case is :
5. A string AB of length 5 m and linear mass density 1 kg m^{-1} is clamped at both the ends with a tension of 144 N. Find minimum frequency (in Hz) of transverse standing wave in string so that a node appears at a distance 3 m from end A.

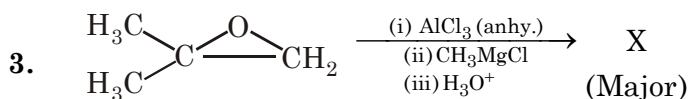
Space for Rough Work

PART-2 : CHEMISTRY

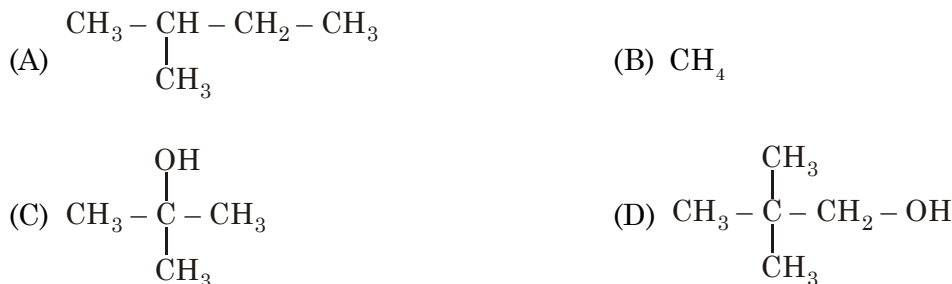
SECTION-I(i) : (Maximum Marks : 15)

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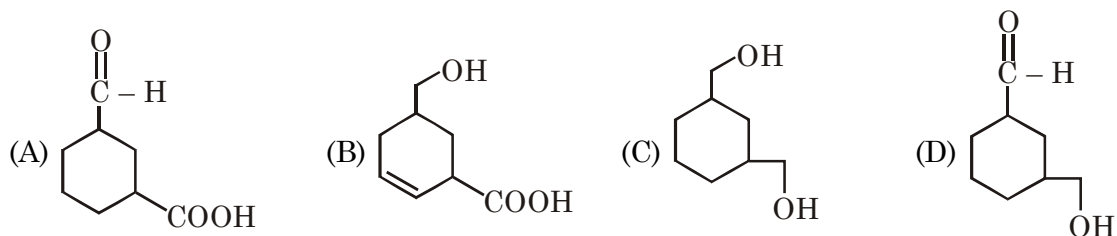
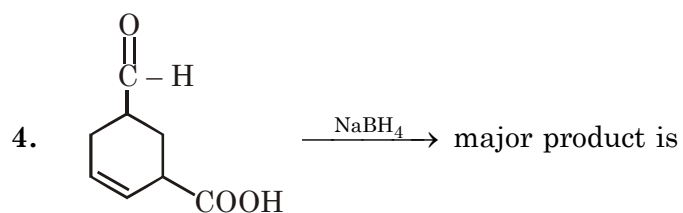
- Which of the following ore does not contain Mg.
 (A) Magnetite (B) Epsom (C) Dolomite (D) Carnalite
- In which of the following reaction SO_2 gas is not produced?



Correct structure of X?



Space for Rough Work



5. For which of the following, ΔG_f° is non-zero ?

- (A) $\text{D}_2(\text{g})$ (B) $\text{Sn}(\text{s}, \alpha)$ (C) $\text{Br}_2(\text{l})$ (D) $\text{Sn}(\text{s}, \beta)$

Space for Rough Work

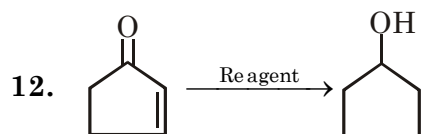
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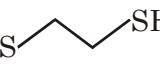
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6. Select correct statements -
- (A) When equal moles of two different ideal gases are mixed at same T & P, then entropy change of mixing will be maximum for the given total moles.
- (B) When different moles of two same ideal gases mixed at same T & P, then entropy change of mixing will be zero
- (C) Two vanderwaal gases are expanded isothermally from V_1 to V_2 , then ΔS for CO_2 will be greater than He
- (D) When two vanderwaal gases are expanded isothermally from V_1 to V_2 , then ΔU for CO_2 will be greater than He.
7. Select the correct statements for water heated in closed rigid vessel. When temperature increase gradually from 25°C then -
- (A) Density of vapour phase decreases
- (B) Density of liquid phase decreases
- (C) A condition will be reached when free vapourisation throughout the liquid occur
- (D) A condition will be reached when liquid passes into gaseous state imperceptibly.

Space for Rough Work

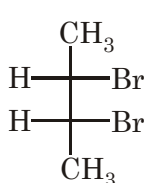
8. Select correct statements -
 (A) Boundry surface diagrams gives the shape of an orbital
 (B) $|\Psi|^2$ is constant on a boundry surface diagram
 (C) Probability density variation can be visualised in boundry surface diagram.
 (D) The boundry surface for $|\Psi|^2$ & $|\Psi|$ are identical.
9. Which of the following statement(s) is/are true?
 (A) Silver is purified by cupellation method
 (B) Tin is purified by liquation followed by poling
 (C) Bessemerisation is used in the metallurgy of copper
 (D) Extraction of Ag by Parke's method is based on distribution law.
10. Which of the following combination/s of species can evolve $O_2(g)$.
 (A) $PbO_2 + \text{warm conc. } H_2SO_4$ (B) $MnO_4^- / OH^- + H_2O_2$
 (C) $XeF_2 + H_2O$ (D) $XeF_6 + H_2O$
11. Which of the following compound do/does not liberate any gas on warming with dil. HCl?
 (A) CaC_2O_4 (B) ZnS (C) FeS (D) K_2SO_4



Select correct reagent(s) for given reaction?

- (A) H_2 / Ni (B) (i)  $/ H^+$ (ii) H_2 / Ni
 (C) DIBALH (D) (i) $BH_3 / T.H.F.$ (ii) CH_3COOH
13. $CH_3 - CCl_3 \xrightarrow[\Delta]{Ag} X \xrightarrow[CaCO_3]{H_2 - Pd} Y \xrightarrow{Br_2 / CCl_4} Z$

Select correct option(s) for given reaction sequence?

- (A) X is $CH_3 - C \equiv C - CH_3$ (B) Y is 
 (C) Y is  (D) Z is 

Space for Rough Work

SECTION-II : (Maximum Marks: 15)

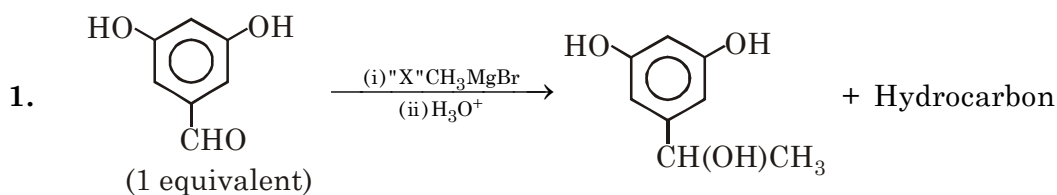
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For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
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2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

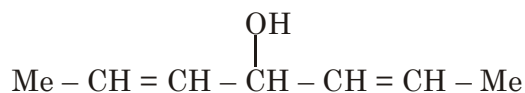
-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.



Find out value of (X)?

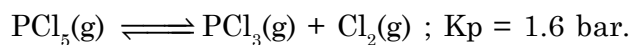
2. Identify total number of stereoisomers for following compounds?



3. How many following metals can evolve NO_(g) with dil. HNO₃.
 Hg, Cu, Pb, Zn, Fe, Al, Ag, Au, Mn

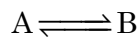
Space for Rough Work

4. 0.1 mole of $\text{PCl}_5(\text{g})$ is introduced in vessel containing a liquid solvent, ($M_w = 160$). At 523K $\text{PCl}_5(\text{g})$ is 50% dissociated & total pressure is found 5.4 bar, at equilibrium. Calculate the minimum weight of solvent in vessel (in gm).



Assume solvent does not dissolve any gas.

5. Following observation is made for reversible reaction first order opposed by first order.



Time(hr)	0	1	∞
Mole % of A	100	80	40

Determine mol % of A after 2 hours

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The position vector of point of intersection of the planes $\vec{r} \cdot (\hat{n}_2 \times \hat{n}_3) = 1$, $\vec{r} \cdot (\hat{n}_3 \times \hat{n}_1) = 2$ and $\vec{r} \cdot (\hat{n}_1 \times \hat{n}_2) = 3$, ($[\hat{n}_1 \hat{n}_2 \hat{n}_3] \neq 0$) is

(A) $\frac{\hat{n}_1 + 2\hat{n}_2 + 3\hat{n}_3}{[\hat{n}_1 \hat{n}_2 \hat{n}_3]}$

(B) $\frac{\hat{n}_1 + 2\hat{n}_2 + 3\hat{n}_3}{[\hat{n}_1 \hat{n}_2 \hat{n}_3]^2}$

(C) $\frac{\hat{n}_2 + \hat{n}_3 + 2\hat{n}_1 + \hat{n}_1 + 3\hat{n}_1 \times \hat{n}_2}{[\hat{n}_1 \hat{n}_2 \hat{n}_3]}$

(D) $\frac{\hat{n}_2 + \hat{n}_3 + 2\hat{n}_1 + \hat{n}_1 + 3\hat{n}_1 \times \hat{n}_2}{[\hat{n}_1 \hat{n}_2 \hat{n}_3]^2}$

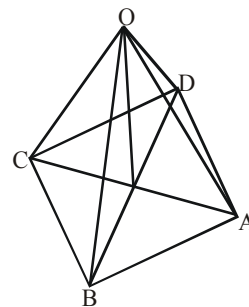
2. Pyramid OABCD has square base ABCD. Edges OA, OB, OC and OD are congruent and $\angle AOB = 45^\circ$. The angle between face OAC and OBD is

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(C) $\frac{2\pi}{3}$

(D) $\frac{5\pi}{6}$



Space for Rough Work

3. Let A,B,C,D be 4 non-coplanar points in space such that $AB = CD$, $AC = BD$ and $AD = BC$. Then the value of $\text{sgn}(\overrightarrow{DA} \cdot \overrightarrow{DB}) + \text{sgn}(\overrightarrow{DB} \cdot \overrightarrow{DC}) + \text{sgn}(\overrightarrow{DA} \cdot \overrightarrow{DC})$ is {sgn is signum function}
- (A) 0 (B) 1 (C) 2 (D) 3
4. The value of $\lim_{x \rightarrow \infty} x^2 \tan \left(\ln \left(\sqrt{\cos \frac{\pi}{x}} \right) \right)$ is equal to
- (A) $\frac{-\pi^2}{2}$ (B) $\frac{\pi^2}{2}$ (C) $\frac{-\pi^2}{4}$ (D) $\frac{\pi^2}{4}$
5. A cuboid is completely inscribed in a hemisphere of radius $\sqrt{2}$ units such that a face 'F' of cuboid completely lies on the base of hemisphere. If one of the sides of 'F' is 2 units, then maximum volume of cuboid is
- (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

6. If $I_n = \int_0^{\pi/2} x (\sin x + \cos x)^n dx$; then-

(A) $\frac{101 I_{101} - \pi/2}{I_{99}} = 200$

(B) $\frac{4 I_4 - \pi/2}{I_2} = 6$

(C) $\frac{101 I_{101} - \pi/2}{I_{99}} = 201$

(D) $\frac{4 I_4 - \pi/2}{I_2} = 8$

7. For the function $f(x) = (x - 1) |x - 3| - 4x + 12$, which of the following holds good ?
- (A) Number of critical points on the graph of $y = f(x)$ is 3.
 (B) Number of points of local extrema of $f(x)$ are 2.
 (C) Number of points of local extrema of $f(x)$ are 3.
 (D) Range of $f(x)$ is $\mathbb{R}$.

Space for Rough Work

8. A variable line moves in a plane such that the product of perpendiculars from $A(a,0)$ & $B(0,0)$ is equal to λ^2 , A and B being on same side of variable line. The locus of foot of perpendicular from $(0,0)$ upon the variable line is a circle which has-
- (A) Radius equal to $\sqrt{\frac{1}{4}a^2 + \lambda^2}$ (B) Centre at $\left(\frac{a}{2}, 0\right)$
- (C) Radius equal to $\sqrt{\frac{1}{4}a^2 - \lambda^2}$ (D) Centre at $\left(0, \frac{a}{2}\right)$
9. Let B be an invertible square matrix and it is the adjoint of matrix A such that $AB = B^T$, then-
- (A) A is an identity matrix. (B) B is a symmetric matrix.
 (C) A is not an identity matrix. (D) B is a skew symmetric matrix.
10. Let A be a real square matrix of order two such that $A^T - I = A^T A$, then -
- (A) A is symmetric matrix (B) $I + A^2 + A^3 + \dots + A^7 + A^8 + A^{10} = O$
 (C) $A^{-1} = I - A$ (D) $\det(A - I) = -1$
 (where $\text{tr}(X)$ denotes trace of a matrix X , X^T denotes transpose of X and I denotes unit matrix)
11. If $y = y(x)$, $y(x) \in \left[0, \frac{\pi}{2}\right)$ is solution of $\left[\frac{1 + 2 \sin x \cos^2(x+y)}{\cos^2(x+y)}\right] dx + \left[\frac{1}{\cos^2(x+y)} + 2 \sec^2 y\right] dy = 0$,
 $y(0) = \frac{\pi}{4}$, then which is/are true -
- (A) $y\left(\frac{\pi}{2}\right) = \frac{\pi}{4}$
- (B) $y\left(\frac{\pi}{4}\right) = \frac{\pi}{4}$
- (C) Equation of tangent to $y = y(x)$ at $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$ is $8x + 12y = 7\pi$
- (D) Equation of tangent to $y = y(x)$ at $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$ is $12x + 8y = 7\pi$

Space for Rough Work

12. If a, b, c are roots of $x^3 + x^2 - 4 = 0$ and $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$, then

(A) A is non-singular

(B) A is orthogonal

(C) $AA^T = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$

(D) $A^{-1} = A^T$

13. $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-zero & non-coplanar vectors with fixed magnitudes. Angles between any of the vector with the normal of the plane containing other two is α , ($\alpha \in (0, \pi)$). Volume of parallelepiped formed is V and its surface area is S . Which of the following is/are true ?

(A) $\frac{V}{S} = \frac{|\cos \alpha|}{2} \left[\frac{1}{|\vec{a}|} + \frac{1}{|\vec{b}|} + \frac{1}{|\vec{c}|} \right]^{-1}$

(B) $V < |\vec{a}||\vec{b}||\vec{c}|$

(C) $\vec{a}, \vec{b}$ and $\vec{c}$ are equally inclined to each other

(D) if angle between $\vec{a}, \vec{b}$ is θ , then angle between $\vec{c}$ and plane of $\vec{a}$ and $\vec{b}$ is $(90^\circ - \theta)$

Space for Rough Work

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. The curve $y = e^{2x} - 18x + 15$ crosses the y-axis at P and has point of minima at Q. If the area bounded by the curve and chord PQ is (use $\ln 3 = 1.1$)
2. Point P(a, b, c) satisfies $a - b + c = 3$ and $a^2 + b^2 + c^2 = 4$. The point P traces a plane figure whose area is
3. The volume enclosed by system of planes given by $|x| + |y| + |z| = 2$ is equal to

Space for Rough Work

4. Line $x + y = 7$ cuts the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ (with centre C) at A and B. If P is a point on AB such that $\angle CPA = \frac{\pi}{4}$, then PA + PB is (take $\sqrt{2} = 1.41$)
5. If $f(x) = \max\{\min(x, x^2), \max(x, x^2)\}$, where $x \geq 0$ then $\frac{d}{dx}((f \circ f)^{-1}(x))$ at $x = 16$ is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	5	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	15
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	32
II	Numerical Value Type (Up to second decimal place)	5	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	15

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator